Bihar Engineering University, Patna

B. Tech. Ist Semester Examination, 2024

Course: B. Tech.
Code: 100102

Subject: Engineering Mathematics –I

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option the following (Any seven question only): **[2 x 7 = 14]**

(a) Which operation is not an elementary row operation?
- (i) Swapping two rows
- (ii) Adding a multiple of one row to another
- (iii) Multiplying a row by zero
- (iv) Multiplying a row by a non-zero constant

(b) The rank of a matrix is:
- (i) Number of non-zero rows in echelon form
- (ii) Number of columns
- (iii) Number of rows
- (iv) None of the above

(c) The inverse of a matrix using Gauss-Jordan method involves:
- (i) Only determinant calculation
- (ii) Augmenting with identity matrix and applying row operations
- (iii) Reducing matrix to diagonal form
- (iv) None of the above

(d) A matrix is diagonalizable if:
- (i) It has repeated eigenvalues only
- (ii) It is symmetric
- (iii) It has a full set of linearly independent eigenvectors
- (iv) It is square

(e) A matrix is similar to a diagonal matrix if:
- (i) It has complex entries
- (ii) It has n linearly independent eigenvectors
- (iii) It is invertible
- (iv) None of the above

(f) A function is Riemann integrable if:
- (i) It is continuous
- (ii) It has infinite discontinuities
- (iii) It is discontinuous everywhere
- (iv) None of the above

(g) Jacobian is:
- (i) A scalar
- (ii) A matrix of partial derivatives
- (iii) A vector
- (iv) A function

(h) Taylor's expansion for multivariable function includes:
- (i) Only first order terms
- (ii) All orders
- (iii) Only second order
- (iv) Only constant term

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(i) Change of variables in double integral involves:

(i) Matrix algebra
(ii) Laplace transforms
(iii) Jacobian
(iv) Only constant term

(j) Triple integral over a region gives:

(i) Area
(ii) Length
(iii) Volume
(iv) Angle

Q.2 (a) What is the difference between a vector space and a subspace? Illustrate with examples. [7]

(b) Define and explain rank, row space, and column space of a matrix with examples. [7]

Q.3 (a) Explain the Cayley-Hamilton Theorem and verify it for a 2x2 matrix. [7]

(b) Define and compute the Jacobian for transformation $x = u + v$, $y = u - v$. [7]

Q.4 (a) Explain the Gauss-Jordan method for finding the inverse of a matrix with an example. [7]

(b) Define Hermitian, Skew-Hermitian, and Unitary matrices. Provide examples for each. [7]

Q.5 (a) Explain and prove the Rank-Nullity theorem. [7]

(b) Describe the method of finding eigenvalues and eigenvectors of a matrix. [7]

Q.6 (a) State and prove Rolle's Theorem with a graphical example. [7]

(b) Define Beta and Gamma functions. Derive the relation between them. [7]

Q.7 (a) Derive the Taylor series for a function of two variables. [7]

(b) Solve a maxima-minima problem using second derivative test. [7]

Q.8 (a) Evaluate a double integral to find area under a curve. [7]

(b) Convert a double integral into polar coordinates and evaluate. [7]

Q.9 **Write Short notes on any two of the following:** **[7 x 2 = 14]**

(a) Properties of Eigen vectors
(b) Riemann Integration & Riemann Sum
(c) Application to area and volume using double and triple integral
(d) Scalar and vector fields

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